

Technical Report: Data Understanding and Measurement Basics

Task 3

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Task 3 – Software/ML Domain

Participant Name: Tanya Misra

Roll Number: 251090052106

Domain: Software/ML

Repository Link: https://github.com/TanTheta017/Synergy_TP

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1. Introduction

Data analysis begins not with code or visualization, but with understanding what a dataset represents. Every value in a table carries meaning only when connected to its variable name, measurement unit, experiment condition, and domain context. This report explains the fundamental concepts required before any formal analysis: observations versus variables, categorical versus numerical data, raw versus processed data, measurement units, and basic summary statistics. Special emphasis is placed on domain-specific examples from Biochemistry, Electronics, and Mechanical engineering to illustrate why context matters. The objective is to build a theoretical foundation that ensures data is interpreted correctly before applying statistical or machine learning methods.

2. Difference Between an Observation and a Variable

An observation is a single record or instance in a dataset. It represents one entity, event, or measurement taken at a specific point in time. A variable is a characteristic or attribute that is measured or recorded for each observation. In tabular terms, observations are rows, and variables are columns. For example, in a dataset of patient blood tests, one patient's test result is an observation, while "glucose level" is a variable.

3. Difference Between Categorical and Numerical Data

Categorical data represents qualities or categories that cannot be measured numerically, such as material type or experiment condition. Numerical data represents measurable quantities, such as temperature or voltage. Categorical data can be nominal (no order) or ordinal (ordered categories). Numerical data can be discrete (countable) or continuous (any value within a range).

4. Domain Examples of Categorical Data

- **Biochemistry:** Experiment condition (control or heated)
- **Electronics:** Load type (low load or high load)
- **Mechanical:** Material type (A or B)

5. Domain Examples of Numerical Data

- **Biochemistry:** pH value
- **Electronics:** Voltage (volts)
- **Mechanical:** Force (newtons)

6. Difference Between Raw Data and Processed Data

Raw data is unprocessed, original data collected directly from instruments or experiments. It may contain errors, missing values, or noise. Processed data has been cleaned, transformed, normalized, or aggregated to make it suitable for analysis. For example, raw voltage readings from a sensor are raw data; the average voltage over one minute is processed data.

7. What Is a Measurement Unit and Why Does It Matter?

A measurement unit is a standard quantity used to express a physical measurement, such as meters, volts, or grams. Units matter because they give meaning to numerical values. The value of "5" has no meaning without its unit (5 volts, 5 meters, 5 grams). Units also allow reproducibility and comparison across experiments.

8. Why Is It Dangerous to Compare Values from Different Units Directly?

Comparing values from different units directly is dangerous because the numerical magnitude does not reflect the actual physical quantity. For example, comparing 5 volts to 5 amperes is meaningless because they measure different properties. Even when measuring the same property, different scales (e.g., meters versus millimeters) can lead to incorrect conclusions if conversion is not applied.

9. Mean, Median, Minimum, Maximum, and Range

- **Mean:** The arithmetic average of all values.
- **Median:** The middle value when data is sorted in order.
- **Minimum:** The smallest value in the dataset.
- **Maximum:** The largest value in the dataset.
- **Range:** The difference between the maximum and minimum values.

10. When Can the Mean Be Misleading?

The mean can be misleading when the data contains outliers (extreme values) or when the distribution is highly skewed. For example, if most measurements are around 10 but one outlier is 1000, the mean will be much higher than the typical value. In such cases, the median is a better measure of central tendency.

11. What Should Be Inspected Before Doing Any Analysis on a Dataset?

Before analysis, one should inspect:

- *The number of observations and variables.*
- *Data types of each variable (categorical or numerical).*
- *Missing values and how they are represented.*
- *Measurement units for all numerical variables.*
- *Outliers or extreme values.*
- *Basic summary statistics (mean, median, min, max, range).*
- *Whether the data is raw or processed.*
- *Any documentation or metadata describing how data was collected.*

12. Questions to Ask Before Analyzing Domain Data from Another Team

- What is the measurement unit for each numerical variable?
- Was the data collected under controlled or real-world conditions?
- Is this raw data or has it been preprocessed?
- What is the meaning of each categorical label?
- Are there any known errors or missing values?
- What instrument or method was used to collect the data?
- What is the time period and sampling frequency?

13. Domain Thinking: Why Unit and Context Matter

A dataset is not just a table of numbers. Each value becomes meaningful only when connected to its variable name, measurement unit, experiment condition, and domain context. For example, pH 7 and 7 volts are not comparable because they measure different physical phenomena. Similarly, 120 N in mechanical engineering refers to force, not a score or percentage. Without domain context, data analysis can lead to incorrect conclusions.

14. Required Domain Examples Table

Table 1: Domain examples showing categorical data, numerical data, and why unit/context matters.

Table 1: Domain Examples

Domain	Categorical Example	Numerical Example	Why Unit/Context Matters
Biochemistry	Experimental condition (Control/Heated)	pH, concentration (mol/L), absorbance	A pH value of 7 and a voltage of 7 V represent completely different quantities and cannot be compared.
Electronics	Load type (Low load/High load)	Voltage (V), current (mA), resistance (Ω)	A value of 5 V measures electrical potential, whereas 5 mA measures electric current. They describe different physical properties.
Mechanical	Material type (Steel/Aluminium)	Force (N), strain (%), displacement (mm)	A force of 120 N indicates the amount of applied force and should not be interpreted as a score or percentage.

15. References

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